AI Assisted Coding Assignment- 8.2

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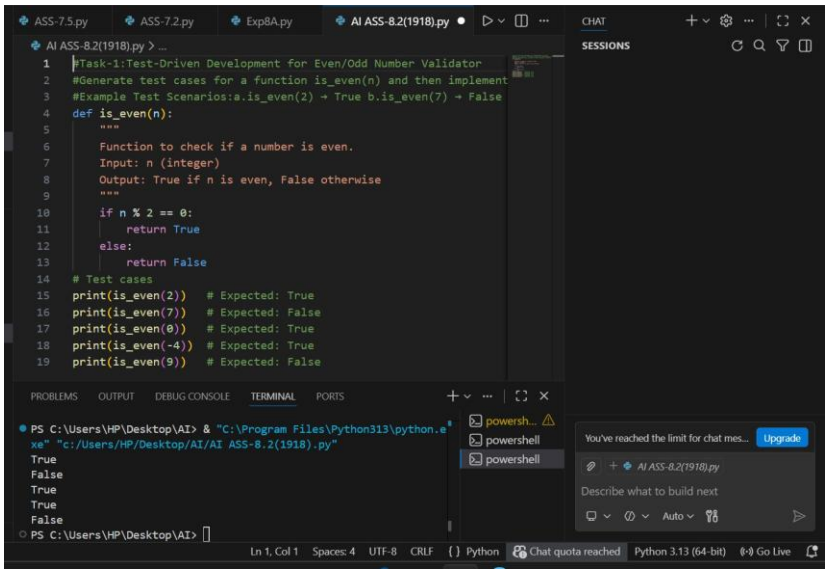
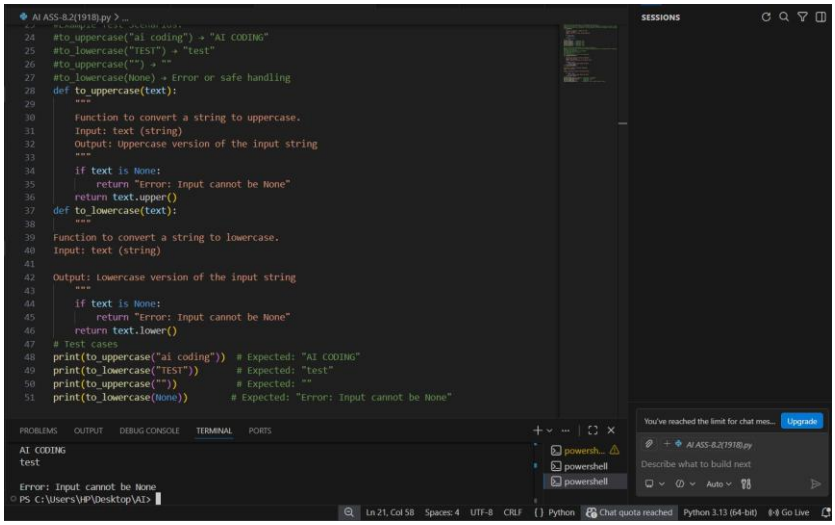
BATCH 30

Task 1 Test-Driven Development for Even/Odd Number Validator
• Use AI tools to first generate test cases for a function is_even(n) and then implement the function so that it satisfies all generated tests.
Requirements:
• Input must be an integer
• Handle zero, negative numbers, and large integers
Example Test Scenarios:
is_even(2 → True
is_even(7 → False
is_even(0 → True
is_even(-4) → True
is_even(9 → False
Expected Output 1
• A correctly implemented is_even() function that passes all AI generated test cases

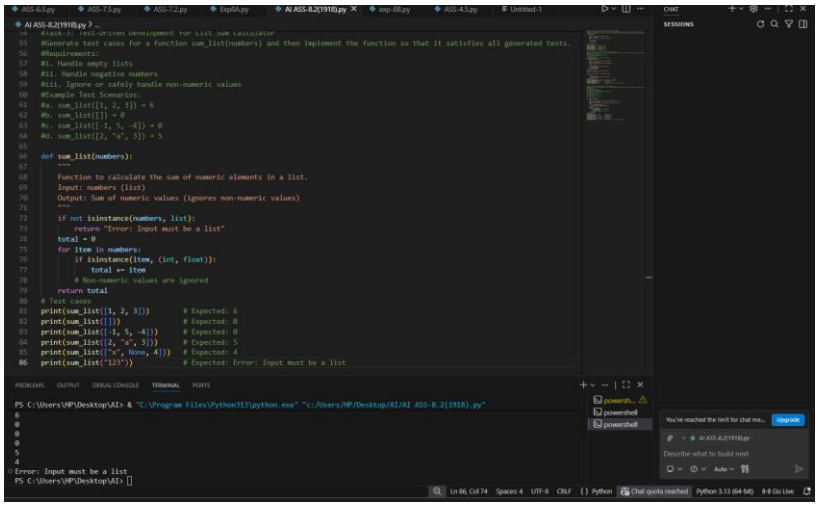


Task 2 Test-Driven Development for String Case Converter
• Ask AI to generate test cases for two functions:
• to_uppercase(text)
• to_lowercase(text)
Requirements:
• Handle empty strings
• Handle mixed-case input
• Handle invalid inputs such as numbers or None
Example Test Scenarios:
to_uppercase("ai coding") → "AI CODING"
to_lowercase("TEST") → "test"
to_uppercase("") → ""

to_lowercase(None) → Error or safe handling
Expected Output 2
• Two string conversion functions that pass all AI-generated test cases with safe input handling.



Task 3 Test-Driven Development for List Sum Calculator
• Use AI to generate test cases for a function sum_list(numbers) that calculates the sum of list elements.
Requirements:
• Handle empty lists
• Handle negative numbers
• Ignore or safely handle non-numeric values
Example Test Scenarios:
sum_list([1, 2, 3 → 6
sum_list([]) → 0
sum_list 1, 5, 4 → 0
sum_list([2, "a", 3 → 5
Expected Output 3
• A robust list-sum function validated using AI-generated test cases.



Task 4 Test Cases for Student Result Class

• Generate test cases for a StudentResult class with the following methods:

• add_marks(mark)

• calculate_average()

• get_result()

Requirements:

• Marks must be between 0 and 100

• Average ≥ 40 → Pass, otherwise Fail

Example Test Scenarios:

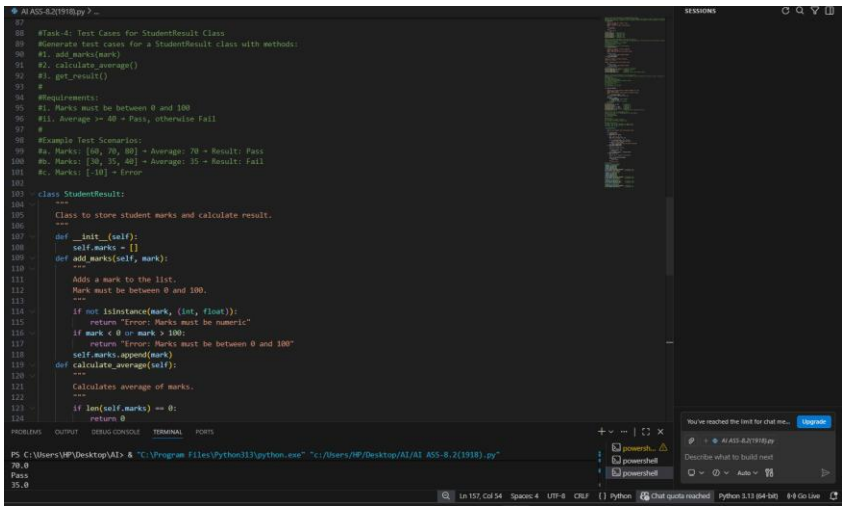
Marks: 60, 70, 80 → Average: 70 → Result: Pass

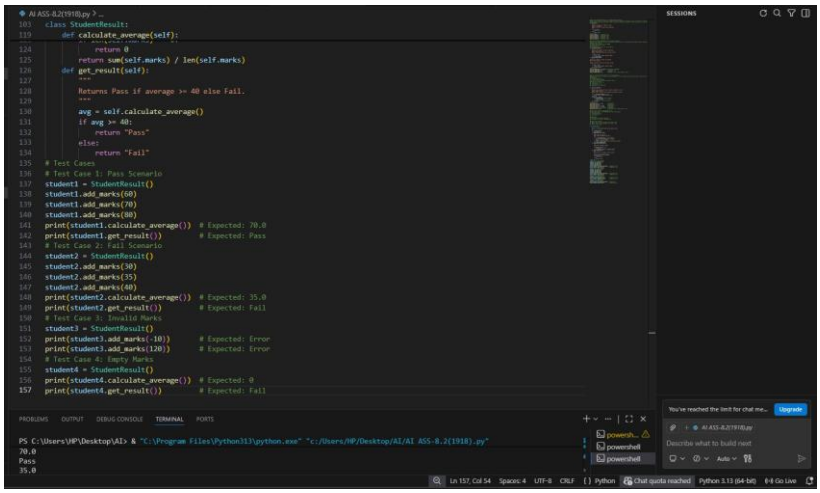
Marks: 30, 35, 40 → Average: 35 → Result: Fail

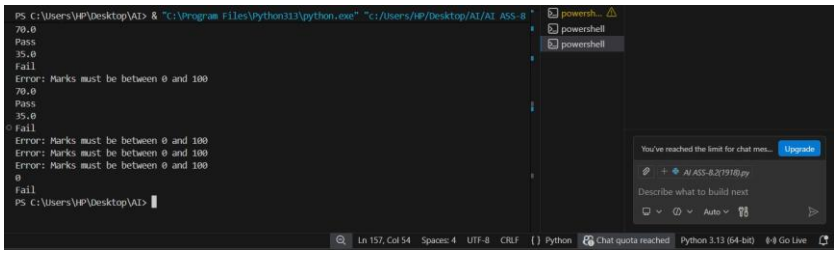
Marks: 10 → Error

Expected Output 4

• A fully functional StudentResult class that passes all AI generated test







Task 5 Test-Driven Development for Username Validator

Requirements:

- Minimum length: 5 characters
- No spaces allowed
- Only alphanumeric characters

Example Test Scenarios:

is_valid_username("user01") → True

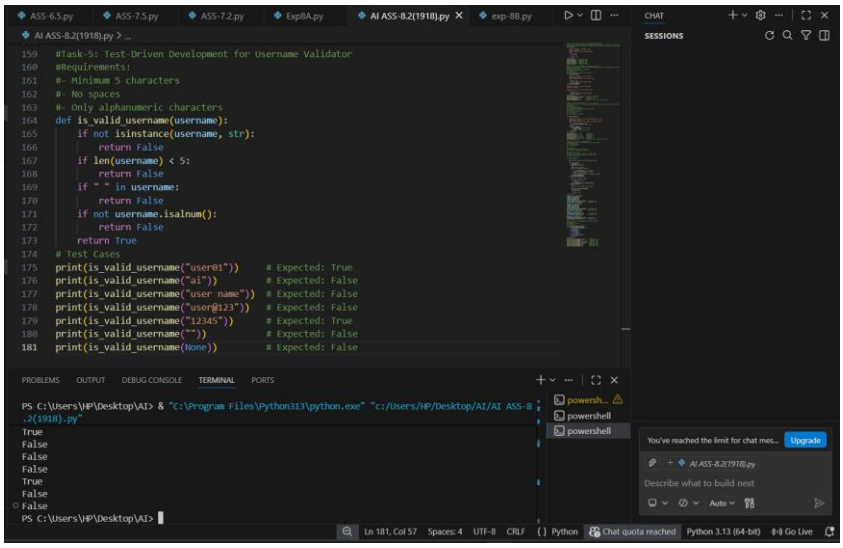
is_valid_username("ai") → False

is_valid_username("user name") → False

is_valid_username("user@123") → False

Expected Output 5

A username validation function that passes all AI-generated test cases.



DOCTEST

python -m doctest -v Ass_8_2.py

True

False

True

True

False

AI CODING

test

Error: Input cannot be None

6

0

0

5
Average: 70.0
Result: Pass
Average: 35.0
Result: Fail
True
False
False
False
All tests passed successfully.
Trying:
is_even(2
Expecting:
True
ok
Trying:
is_even(7
Expecting:
False
ok
Trying:
is_even(0
Expecting:
True
ok
Trying:
is_even(-4)
Expecting:
True
ok
Trying:
is_even(9
Expecting:
False
ok
10 items had no tests:
Ass_8_2

Ass_8_2.StudentResult
Ass_8_2.StudentResult.init
Ass_8_2.StudentResult.add_marks
Ass_8_2.StudentResult.calculate_average
Ass_8_2.StudentResult.get_result
Ass_8_2.is_valid_username
Ass_8_2.sum_list
Ass_8_2.to_lowercase
Ass_8_2.to_uppercase
1 items passed all tests:
5 tests in Ass_8_2.is_even
5 tests in 11 items.
5 passed and 0 failed.
Test passed.

import pytest
from Ass_8_2 import is_even, to_uppercase, to_lowercase, sum_list, StudentResult, is_vali
d_username

Test cases for Task 1 - Even/Odd Number Validator

def test_is_even():
assert is_even(2 == True
assert is_even(7 == False
assert is_even(0 == True

```
assert is_even(-4) == True
assert is_even(9  == False
```

Test cases for Task 2 - String Case Converter

```
def test_to_uppercase():
assert to_uppercase("ai coding") == "AI CODING"
assert to_uppercase("") == ""
assert to_uppercase(None) == "Error: Input cannot be None"
assert to_uppercase("Test") == "TEST"
def test_to_lowercase():
assert to_lowercase("TEST") == "test"
assert to_lowercase("") == ""
assert to_lowercase(None) == "Error: Input cannot be None"
assert to_lowercase("Test") == "test"
```

Test cases for Task 3 - List Sum Calculator

```
def test_sum_list():
assert sum_list([1, 2, 3   == 6
assert sum_list([]) == 0
assert sum_list   1, 5,   4   == 0
assert sum_list([2, "a", 3   == 5
assert sum_list([2, 3,  3, "a", 4   == 6
assert sum_list([100,   50, 20   == 70
```

Test cases for Task 4 - StudentResult Class

```
def test_student_result():
student = StudentResult()
student.add_marks(60)
student.add_marks(70)
student.add_marks(80)
assert student.calculate_average() == 70.0
assert student.get_result() == "Pass"
student = StudentResult()
student.add_marks(30)
student.add_marks(35)
student.add_marks(40)
assert student.calculate_average() == 35.0
assert student.get_result() == "Fail"
```

Test cases for Task 5 - Username Validator

```
def test_is_valid_username():
assert is_valid_username("user01") == True
assert is_valid_username("ai") == False
assert is_valid_username("user name") == False
assert is_valid_username("user@123") == False
assert is_valid_username("validUser") == True
assert is_valid_username("us") == False
```

platform win32 -- Python 3.12.3, pytest-9.0.2, pluggy-1.6.0 rootdir: PS C:\Users\HP\Desktop\AI>

A.I.AC

rootdir: PS C:\Users\HP\Desktop\AI> A.I.AC
collected 6 items
Ass_8_2.py ......
[100%]
==========

6 passed in 0.09s